Lab Assignment

Q1: In modern operating systems, job scheduling is crucial for ensuring that critical system tasks are executed in a timely manner. Some processes, like handling interrupts or urgent system maintenance tasks, need higher priority than regular user processes. Design and implement a Priority Queue in C to manage system processes based on their priority levels. Higher priority processes should be executed first, while processes with the same priority are handled in a first-come, first-served (FIFO) manner. Implement the system with at least 10 processes, each having a unique ID and a priority value (10 being the highest, 1 being the lowest). Some examples of processes are: Interrupt Handling, System Maintenance, User Applications, and Background Tasks.

Your program should support the following operations:

- Insert Process: Add a new process to the queue with a priority level and process ID.
- Remove Highest Priority Process: Remove and return the process with the highest priority for execution.
- Peek at Highest Priority Process: View the process with the highest priority without removing it.
- Change Process Priority: Modify the priority of a process already in the queue.

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Q2: In real-world applications such as data storage, file transfer, and communication systems, efficient data compression is critical for reducing storage space and bandwidth usage. One of the most widely used algorithms for lossless data compression is Huffman Coding, which is based on a greedy approach to minimize the average length of encoding symbols. Design and implement a Huffman Coding algorithm in C to compress a given text file. Your implementation should build a binary tree based on the frequency of characters in the input, and then generate an optimal prefix-free binary code for each character to achieve maximum compression. Your program should be able to read from a text file of any size. For demonstration, use a sample text file containing English sentences or paragraphs.

Output: Display the original text, the Huffman codes for each character, and the compressed binary string. Compare and display the compression ratio.

Q3: You are working as a software engineer in a company that manages conference room scheduling. Multiple meetings are scheduled throughout the day, and your goal is to book the maximum number of non-overlapping meetings in a single room. Implement a Greedy Algorithm in C to solve this problem**.** The program should handle at least **10 jobs** with different start and end times.

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